STAT 2793 HOMEWORK 6 SOLUTIONS

1a) <u>STEP 1</u> LET μ_N REPRESENT THE MEAN BREASTMILK VOLUME OF NONSMOKING MOTHERS (POPULATION 1)

LET μ_S REPRESENT THE MEAN BREASTMILK VOLUME OF SMOKING MOTHERS (POPULATION 2)

$H_0: \mu_N - \mu_S = 0$

$H_a: \mu_N - \mu_S > 0$

<u>STEP 2</u>: $\alpha = 0.05$

<u>STEP 3</u>: USE MANN WHITNEY WILCOXON TEST (SPECIFIED)

WHY? - 2 INDEPENDENT SAMPLES

- SMALL SAMPLE SIZES
- NO UNDERLYING DISTRIBUTION ASSUMPTIONS. OTHER THAN THEY HAVE THE SAME SHAPE.

<u>STEP 4</u>:

H_a: ">" → RIGHT TAILED

$\alpha = 0.05$

USE M → $\left.\begin{array}{l} n_1 = n_N = 6 \\ n_2 = n_S = 7 \end{array}\right\}$ TABLE VI: $M_{6,7,0.05} = 54$.

<u>DECISION RULE</u>: REJECT H_0 IF $M_{CALC} > 54$.

STEP 5 :

SMOKERS	RANK	NON SMOKERS	RANK
545	1	745	5
593	2	850	8
621	3	903	10
655	4	930	11
767	6	947	12
793	7	983	13
895	9		

M_{CALC} = SUM OF POPULATION 1 → NONSMOKERS RANKS

$= 5 + 8 + 10 + 11 + 12 + 13$

$= 59$

SINCE $M_{CALC} = 59 > 54$, WE REJECT H_0.

THERE IS SUFFICIENT EVIDENCE TO SUGGEST NON SMOKING MOTHERS HAVE MORE BREAST MILK THAN SMOKING MOTHERS.

b) WE COULD USE: – 2 INDEPENDENT SAMPLE Z TEST

– 2 INDEPENDENT SAMPLE t – TEST